

CHAPTER #15

"Multiple Integrals"

Exercise #15.1 - Evaluating Iterated Integrals: (1-14)

01)  $\int_1^2 \int_0^4 2ny \, dy \, dn$

Solution:

$$= \int_1^2 2n \int_0^4 y \, dy \, dn$$

$$\int_1^2 2n \left[ \frac{y^2}{2} \right]_0^4 \, dn$$

$$\int_1^2 n \left[ y^2 \right]_0^4 \, dn$$

$$\int_1^2 n [16 - 0] \, dn$$

$$\int_1^2 16n \, dn$$

$$16 \int_1^2 n \, dn$$

$$16 \int_1^2 \frac{n^2}{2} \, dn$$

$$\frac{16}{8} \left[ \frac{n^3}{3} \right]_1^2$$

$$2 \left[ n^3 \right]_1^2$$

$$2 [8 - 1]$$

$$2 [7]$$

$$[14]$$

Answer

02)  $\int_0^2 \int_{-1}^1 (n-y) \, dy \, dn$

Solution:-

$$= \int_0^2 \left[ \int_{-1}^1 n \, dy - \int_{-1}^1 y \, dy \right] \, dn$$

$$\int_0^2 \left[ n \int_{-1}^1 1 \, dy - \int_{-1}^1 y \, dy \right] \, dn$$

$$\int_0^2 \left[ n \cdot y \Big|_{-1}^1 - \frac{y^2}{2} \Big|_{-1}^1 \right] \, dn$$

$$\int_0^2 \left[ n [1 - (-1)] - \frac{1}{2} [1 - 1] \right] \, dn$$

$$\int_0^2 \left[ n [2] - \frac{1}{2} [0] \right] \, dn$$

$$\int_0^2 [2n - 0] \, dn$$

$$\int_0^2 2n \, dn$$

$$2 \int_0^2 n \, dn$$

$$2 \left[ \frac{n^2}{2} \right]_0^2$$

$$n^2 \Big|_0^2$$

$$(4 - 0)$$

$$[4] \text{ Answer}$$

$$03) \int_{-1}^0 \int_{-1}^1 (x+y+1) dx dy.$$

Solution:

Constant Treat.

$$= \int_{-1}^0 \left[ \int_{-1}^1 x dx + \int_{-1}^1 y dx + \int_{-1}^1 1 dx \right] dy.$$

$$\int_{-1}^0 \left[ \frac{x^2}{2} \Big|_{-1}^1 + y [x]_{-1}^1 + [x]_{-1}^1 \right] dy.$$

$$\int_{-1}^0 \left[ \frac{1}{2} [(1-(-1))] + y [1-(-1)] + [1-(-1)] \right] dy.$$

$$\int_{-1}^0 \left[ 0 + 2y + 2 \right] dy$$

$$\int_{-1}^0 2 [y+2] dy \Rightarrow 2 \int_{-1}^0 y dy + \int_{-1}^0 2 dy$$

$$2 \left[ \frac{y^2}{2} \Big|_{-1}^0 + 2 [y]_{-1}^0 \right]$$

$$[0-(1)] + 2 [0-(-1)]$$

$$-1 + 2 (+1)$$

$$-1 + 2$$

$$= 1 \text{ Ans.}$$

$$04) \int_0^1 \int_0^1 \left( 1 - \frac{x^2 + y^2}{2} \right) dx dy.$$

Solution:-

$$= \int_0^1 \int_0^1 \left( 1 - \frac{x^2}{2} - \frac{y^2}{2} \right) dx dy.$$

$$= \int_0^1 \left[ \int_0^1 dx - \frac{1}{2} \int_0^1 x^2 dx - \frac{y^2}{2} \int_0^1 dx \right] dy.$$

$$= \int_0^1 \left[ x \Big|_0^1 - \frac{1}{2} \frac{x^3}{3} \Big|_0^1 - \frac{y^2}{2} x \Big|_0^1 \right] dy.$$

$$\int_0^1 \left[ (1-0) - \frac{1}{6} (1-0) - \frac{y^2}{2} (1-0) \right] dy$$

$$\int_0^1 \left[ 1 - \frac{1}{6} - \frac{y^2}{2} \right] dy.$$

$$= \int_0^1 dy - \frac{1}{6} \int_0^1 dy - \frac{1}{2} \int_0^1 y^2 dy$$

$$[y] \Big|_0^1 - \frac{1}{6} [y] \Big|_0^1 - \frac{1}{2} \left[ \frac{y^3}{3} \right] \Big|_0^1$$

$$= 1-0 - \frac{1}{6} [1-0] - \frac{1}{6} [1-0]$$

$$= \frac{1}{1} - \frac{1}{6} - \frac{1}{6} \Rightarrow \frac{6-1-1}{6} \Rightarrow \frac{6-2}{6} = \frac{4}{6} = \frac{2}{3}$$

Answer

$$05) \int_0^3 \int_0^2 (4 - y^2) dy dn$$

Solution:-

$$= \int_0^3 \left[ 4 \int_0^2 dy - \int_0^2 y^2 dy \right] dn$$

$$\int_0^3 \left[ 4 \left[ y \right]_0^2 - \left[ \frac{y^3}{3} \right]_0^2 \right] dn$$

$$\int_0^3 \left[ 4 [2-0] - \frac{1}{3} [8-0] \right] dn$$

$$\int_0^3 \left[ 4 [2] - \frac{1}{3} [8] \right] dn$$

$$\int_0^3 \left[ 8 - \frac{8}{3} \right] dn$$

$$8 \int_0^3 dn - \frac{8}{3} \int_0^3 dn$$

$$8 \left[ [n]_0^3 - \frac{8}{3} [n]_0^3 \right]$$

$$8 \left[ [3-0] - \frac{8}{3} [3-0] \right]$$

$$8 [3] - \frac{8}{3} [3]$$

$$24 - 8$$

$$= 16 \text{ Answer}$$

$$06) \int_0^3 \int_{-2}^0 (n^2 y - 2ny) dy dn$$

Solution:-

$$= \int_0^3 \left[ n^2 \int_{-2}^0 y dy - 2n \int_{-2}^0 y dy \right] dn$$

$$= \int_0^3 \left[ n^2 \left[ \frac{y^2}{2} \right]_{-2}^0 - 2n \left[ \frac{y^2}{2} \right]_{-2}^0 \right] dn$$

$$\int_0^3 \left[ n^2 \cdot \frac{1}{2} [y^2]_{-2}^0 - n [y^2]_{-2}^0 \right] dn$$

$$\int_0^3 \left[ \frac{n^2}{2} [0 - (4)] - n [0 - (4)] \right] dn$$

$$\left[ \frac{n^2}{2} \cdot -\frac{4}{2} - n \cdot -4 \right] dn$$

$$\int_0^3 (-2n^2 + 4n) dn$$

$$= -2 \int_0^3 n^2 dn + 4 \int_0^3 n dn$$

$$= -2 \left[ \frac{n^3}{3} \right]_0^3 + \frac{4}{2} \left[ \frac{n^2}{2} \right]_0^3$$

$$= -\frac{2}{3} [n^3]_0^3 + 2 [n^2]_0^3$$

$$= -\frac{2}{3} [27-0] + 2 [9-0]$$

$$= -\frac{2}{3} [27] + 2 [9]$$

$$= -18 + 18$$

$$= 0 \text{ Answer}$$

When any term is given in  $D(n)$  and it's  $N(n)$  is on the Num then we write it as  $\frac{N(n)}{D(n)}$

Q7)  $\int_0^1 \int_0^1 \frac{y}{1+ny} \, dndy$

Solution:

$$= \int_0^1 \left[ \int_0^1 \frac{y}{1+ny} \, dn \right] dy$$

$$= \int_0^1 \left[ \ln(1+ny) \Big|_0^1 \right] dy$$

$$= \int_0^1 [\ln(1+(1)(y)) - \ln(1+(0)(y))] dy$$

$$\int_0^1 [\ln(1+y) - \underbrace{\ln(1)}_{=0 \text{ calculator}}] dy$$

Substitute:

$$u = 1+y$$

$$du = dy$$

→ ILATE

$$\int_1^2 \frac{1}{u} \ln(u) \, du \rightarrow \text{Integration by Parts}$$

Limit change

$$\frac{y=1}{2}, \frac{y=0}{1}$$

Limit change

$$\ln u \int du - \int \left( \frac{d}{du} \ln u \int 1 \, du \right) du$$

$$\ln u \cdot u \Big|_1^2 - \int_1^2 \frac{1}{u} \cdot u \, du$$

$$u \cdot \ln u \Big|_1^2 - \int_1^2 1 \, du \Big|_1^2$$

$$u \cdot \ln u \Big|_1^2 - u \Big|_1^2$$

$$2 \ln 2 - 1 \ln(1) - [2 - 1]$$

$$2 \ln 2 - 0 - [1]$$

$$\boxed{2 \ln 2 - 1}$$

Answer



$$10) \int_0^1 \int_1^2 n y e^n dy dn$$

$$11) \int_{-1}^2 \int_0^{\pi/2} y \sin n dn dy$$

Solution:

$$= \int_0^1 \left[ n e^n \int_1^2 y dy \right] dn$$

$$= \int_0^1 \left[ n e^n \left[ \frac{y^2}{2} \right]_1^2 \right] dn$$

$$= \int_0^1 \left[ \frac{n e^n}{2} [4-1] \right] dn$$

$$= \int_0^1 \left[ \frac{n e^n}{2} [3] \right] dn$$

$$= \int_0^1 \left[ \frac{3}{2} n e^n \right] dn$$

$$\frac{3}{2} \int_0^1 n e^n dn \xrightarrow{\text{ILATE}} \text{Integration by Parts...}$$

$$\frac{3}{2} \left[ n \int e^n dn - \int \left( \frac{d}{dn} n \right) \int e^n dn \right] dn$$

$$\frac{3}{2} \left[ n e^n - \int 1 e^n dn \right]$$

$$\frac{3}{2} \left[ n e^n \right]_0^1 - \left[ e^n \right]_0^1$$

$$\frac{3}{2} [e^1 - 0] - [e^1 - 1]$$

$$\frac{3}{2} [e - e + 1]$$

$$= \boxed{\frac{3}{2}} \text{ Answer}$$

Solution:-

$$= \int_{-1}^2 y \left[ \int_0^{\pi/2} \sin n dn \right] dy$$

$$= \int_{-1}^2 y \left[ [-\cos n]_0^{\pi/2} \right] dy$$

$$= \int_{-1}^2 -y [\cos \pi/2 - \cos(0)] dy$$

$$\int_{-1}^2 [-y [0-1]] dy$$

$$\int_{-1}^2 y dy$$

$$= \left[ \frac{y^2}{2} \right]_{-1}^2$$

$$\frac{1}{2} [4-1]$$

$$\frac{1}{2} \cdot 3 = \boxed{\frac{3}{2}}$$

Answer.

$$12) \int_{\pi}^{2\pi} \int_0^{\pi} (\sin n + \cos y) dn dy$$

Solution:

$$\int_{\pi}^{2\pi} \left[ \int_0^{\pi} \sin n dn + \cos y \int_0^{\pi} 1 dn \right] dy$$

$$\int_{\pi}^{2\pi} \left[ [-\cos n]_0^{\pi} + \cos y [n]_0^{\pi} \right] dy$$

$$\int_{\pi}^{2\pi} [-\cos(\pi) + \cos(0) + \cos y (\pi - 0)] dy$$

$$\int_{\pi}^{2\pi} [-[-1-1] + \cos y (\pi)] dy$$

$$\int_{\pi}^{2\pi} [-[-2] + \pi \cos y] dy$$

$$\int_{\pi}^{2\pi} [2 + \pi \cos y] dy$$

$$\begin{aligned} & 2 \int_{\pi}^{2\pi} dy + \pi \int_{\pi}^{2\pi} \cos y dy \\ & 2 [y]_{\pi}^{2\pi} + \pi [\sin y]_{\pi}^{2\pi} \\ & 2 (2\pi - \pi) + \pi [\sin 2\pi - \sin \pi] \\ & 2 \pi + \pi [0 - 0] \\ & \boxed{2\pi} \text{ Answer} \end{aligned}$$

When any term's derivative is given in multiplication then we add in that term by 1 and that power in denominator.

$$13) \int_1^4 \int_1^e \frac{\ln n}{ny} dn dy$$

Solution:-

$$= \int_1^4 \left[ \frac{1}{y} \int_1^e \ln n \cdot \frac{1}{n} dn \right] dy$$

Rule 11

$$\int_1^4 \frac{1}{y} \left[ \frac{(\ln n)^2}{2} \right]_1^e dy$$

$$\int_1^4 \frac{1}{2y} \left[ (\ln(e))^2 - (\ln(1))^2 \right] dy$$

$\ln(e) = 1$        $\ln(1) = 0$

$$\int_1^4 \frac{1}{2y} [1 - 0] dy$$

$$\frac{1}{2} \int_1^4 \frac{1}{y} dy$$

When any denominator term integration find  $\Rightarrow$  it's derivative then is given in the numerator, we write that term in  $[\ln]$

$$\frac{1}{2} [\ln y]$$

$$\frac{1}{2} [\ln(4) - \ln(1)]$$

$$\frac{1}{2} \ln(4) - 0$$

$$\boxed{\frac{1}{2} \ln 4} \text{ Answer}$$

$$14) \int_{-1}^2 \int_1^2 n \ln y dy dn$$

Solution:-

$$= \int_{-1}^2 \int_1^2 n \ln y dn dy$$

We can change the order  $dn, dy$  and integrals limits:

through Fubini's Theorem:-

$$= \int_1^2 \int_{-1}^2 n \ln y dn dy \Rightarrow \text{question}$$

$$= \int_1^2 \ln y \left[ \int_{-1}^2 n dn \right] dy$$

$$\int_1^2 \ln y \left[ \frac{n^2}{2} \right]_{-1}^2 dy$$

$$\frac{1}{2} \int_1^2 \ln y [4 - 1] dy$$

$$\frac{1}{2} \int_1^2 3 \ln y dy$$

$$\frac{3}{2} \int_1^2 [\ln y \cdot 1] dy$$

Now, integration by parts

$$\ln y \int 1 dy - \int \left( \frac{d}{dy} \ln y \int 1 dy \right) dy$$

$$\ln y \cdot y - \int \left( \frac{1}{y} \cdot y \right) dy$$

$$y \ln y - \int 1 dy$$

$$y \ln y - y \Big|_1^2$$

Exp. Multiplication

$$[2 \ln(2) - \ln(1)] - [2 - 1]$$

$$\frac{3}{2} [2 \ln 2 - 0 - 1]$$

$$\frac{3 \cdot 2 \cdot \ln 2 - 3}{2}$$

$$\boxed{3 \ln 2 - \frac{3}{2}} \text{ Ans}$$

# Evaluating Double Integrals over Rectangles

15-22, evaluate the double integrals over the given region R.

15)  $\iint_R (6y^2 - 2n) dA$ ,  $R: 0 \leq n \leq 1, 0 \leq y \leq 2$ .

Solution:  
 $\int_0^1 \int_0^2 (6y^2 - 2n) dy dn$

$= \int_0^1 \left[ 6 \int_0^2 y^2 dy - 2n \int_0^2 dy \right] dn$

$\int_0^1 \left[ 6 \cdot \frac{y^3}{3} \Big|_0^2 - 2n [y]_0^2 \right] dn$

$2 \int_0^1 \left[ y^3 \Big|_0^2 - n[y]_0^2 \right] dn$

$2 \left[ \frac{1}{4} [8n] - n [2-0] \right] dn$

$\int_0^1 16 - 2n$

$2 \cdot 16 \int_0^1 dn - 2 \int_0^1 n dn$

$16 [n]_0^1 - \frac{1}{2} \left[ \frac{n^2}{2} \right]_0^1$

$16 [1-0] - \frac{1}{4} [1-0]$

$16 - \frac{1}{4}$   
15 Answer

Ans = 14  
 retn do again...

16)  $\iint_R \left( \frac{\sqrt{n}}{y^2} \right) dA$ ,  $R: 0 \leq n \leq 4, 1 \leq y \leq 2$ .

Solution  
 $\int_0^4 \int_1^2 \left( \frac{\sqrt{n}}{y^2} \right) dy dn$

$\int_0^4 \int_1^2 (y^{-2} \sqrt{n}) dy dn$

$\int_0^4 \left[ \sqrt{n} \int_1^2 (y^{-2} dy) \right] dn$

$\int_0^4 \left[ \sqrt{n} \left[ -\frac{y^{-1}}{-1} \right]_1^2 \right] dn$

$\int_0^4 \left[ -\sqrt{n} \left[ \frac{1}{y} \right]_1^2 \right] dn$

$\int_0^4 \left[ -\sqrt{n} \left[ \frac{1}{2} - \frac{1}{1} \right] \right] dn$

$\int_0^4 -\sqrt{n} \left( \frac{1-2}{2} \right) dn$

$\frac{-\sqrt{n}}{2} \Big|_0^4$

$\int_0^4 \frac{\sqrt{n}}{2} dn$

$\frac{1}{2} \int_0^4 (n)^{1/2} dn$

$\frac{1}{2} \left[ \frac{n^{3/2}}{3/2} \right]_0^4$

$\frac{1}{2} \cdot \frac{(4)^{3/2}}{3/2}$

$\frac{1 \cdot 8}{3}$

8/3 Answer

Ans = 8/3